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CSC 1310 103

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Lab Documentation

1.



b. The buckets began to fill with much more data rather than spreading them out amongst more buckets

c. The lists would become *smaller* as the data is distributed even more

d. I expect searching for data in 6000 buckets would be far faster than 5, as there are more lists that can be transversed quickly rather than searching through 5 massive lists one at a time

e. A hash table with many buckets may use up unnecessary storage when not all of the allotted buckets are filled.

2.

a. The function uses a for loop to iterate through each character in the string and grab it, putting it into a variable. From there, it uses a hash calculation that allows the compiler to distribute the bits uniformly and assigns unique ASCII codes according to its numeric value. The modulo operator makes sure the outcome stays within bounds of the hash table

b. Without the modulo operator, the bounds of the value could exceed the table, causing an error and overflow.

3.

a. Two videos may be assigned to the same bucket due to the modulo operator finding them to have the same index. When the video IDs are converted to ASCII, there is a chance they have the same value from the “remainer” of the modulo assigning them the same bucket.

b. My program handles this situation by implementing chaining and creating a new array if there is a duplicate value in a node. This creates a new linked list attached to the node that holds multiple values that may have the same index attributed to the ASCII.

4.

a. When a new video is added to the table, the whole chain in a bucket is transversed to see if there is a duplicate value. If there is a node with values, the loop continues. When there is a duplicate, the value is replaced with the new key. If there is not a duplicate, the function moves onto the new node in the entry.

b. When there is a duplicate entry, the function locates the duplicate and updates the chain to replace the old duplicate with the new duplicate. There is not a full duplicate that will be available in the final hash table, just a replacement of the old duplicate.

c. We need to be able to ensure no duplicates or null entries are missed. If we do not search the whole list, there is a chance there is a value that could interfere with the new value being put in, causing an error in data handling and storage in the future.

5.

a. Searching would be much more difficult and slow. Inserting, deleting, and doing any sort of editing to the array would be inefficient as every item would have to shift at any edit. Searching would be completely linear in an array, while a hash array uses mapped keys to find the value, making it more efficient.

b. I expect hash tables to be far more efficient, as instead of a strictly linear search algorithm, there is a mapping technique implemented that makes the time efficiency faster and

more reliable for large data sets that can be divided amongst separate buckets to search for instead of just one large list of items.

6.

a. This tells us that on average, out of 6281 videos and 1010 buckets, there are 6.22 videos per bucket. Because the load factor is greater than one, that means there is a possibility of collision and keys are being hashed to the same bucket often. The load factor being this high impacts performance and memory usage as more items are being assigned to the same bucket to be stored and sorted.

7.

a.
```
7.) Exit
CHOOSE 1-7: 5
You chose to Display Hash Table Stats
Unique Videos: 6281
Buckets: 1010
Occupied 1010
Buckets with collision: 999
Longest Chain: 16
Average Entries per Bucket (Load factor): 6.21881
```

b. I believe the hash table is distributing videos reasonably well as every single bucket is occupied with a unique video and most buckets are still being filled with videos. While the load factor is high and collisions are high, out of the amount of buckets, the longest chain is only 16. This means that the bucket with the most collisions only has 16 same indices rather than an imbalanced bucket holding most collisions while other buckets remain nearly empty.